LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to turbine engine starting: Basic circuit of turbo prop engine starting system, turbo starter system, and starter generator system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to turbine engine starting: - Basic circuit of turbo prop engine starting system, turbo starter system, and starter generator system. Safeties to be taken while gas turbine engine starting. Practical on engine starting system.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of engine starting system circuit verification, safety check of starter generator isolation, and verification of circuit protection devices.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main components of a turbo prop starting system?
(b) Name one circuit protection device used in engine starting circuits.
(c) Why is the starter generator used during the starting phase?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic circuit of turbo prop engine starting system • Function of starter generator in starting mode • Role of circuit protection devices and actuators • Sequence of starter engagement and disengagement	L & Ds	PPT, IB	15 min
(b)	• Safeties to be taken while gas turbine engine starting • Pre-start checks for starter generator system • Emergency procedures for failed engine start • Verification of starter system isolation switches	L & Ds	PPT, PDF	20 min
(c)	• Practical on engine starting system circuit verification • Identification of starter generator terminals • Testing of circuit protection devices • Observation of starter engagement sequence	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the function of the starter generator during the cranking phase.
(b) List two critical safety precautions required before initiating engine start.
(c) Describe the role of circuit protection devices in the starting system.

12. Home Assignment and Follow Up	Prepare a brief note on the operation of impulse coupling and booster coils ignition software as covered in LP 19, focusing on their integration with the starter generator system.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Auxiliary starting devices: - Impulse coupling, booster coils ignition s/w and Demonstration of H.E.I.U operation. Safety precautions to be taken while working on high energy /voltage devices.	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min